ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	1304164
ToLID	**icTroFerr3**
Species	Troglocharinus ferreri
Class	Insecta
Order	Coleoptera

Genome Traits	Expected	Observed
Haploid size (bp)	352,998,662	404,204,087
Haploid Number	12 (source: direct)	651
Ploidy	2 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 6.6.Q64

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Haploid Number is different from Expected

. Kmer completeness value is less than 90 for pri
. Assembly length loss > 3% for pri

Curator notes

. Interventions/Gb: None
. Contamination notes: "FCS-GX identified 2766 contigs as being (mostly) bacterial contaminants and these were removed from the assembly to no detriment of the busco statistics or cN50."
. Other observations: "PacBio sequencing was done in Oslo and different specimens were used for long read sequencing and HiFi: icTroFerr3 (HiFi) and icTroFerr4 mixed with icTroFerr5 (Hi-C). Assembly was performed by the CNAG. We first assembled the HiFi reads with Hifiasm and then ran purge_dups on it to collapse duplicated haplotigs and repeats. The Hi-C mapped very poorly to the purged assembly and no meaningful scaffolding was detected in the pretextmap. In fact only 0.3% of the HiC reads generated valid read pairs in Cis (> 1kb) or Trans. More than 60% of HiC reads were tagged as PCR dups. Given the very poor quality of the scaffolding we therefore decided to submit a contig level assembly for this species. For the purpose of this EAR report the pre-curated assembly consisted of the purged contig assembly before decontamination. The curated assembly is simply the contig assembly (3417 contigs) minus the 2766 contigs that we removed as bacterial contaminants. The -haploid number- above just contains the total number of contigs in this assembly"

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	711,250,490	404,204,087
GC %	36.68	33.56
Gaps/Gbp	0	0
Total gap bp	0	0
Scaffolds	3,417	651
Scaffold N50	2,148,059	3,443,960
Scaffold L50	85	34
Scaffold L90	1,033	139
Contigs	3,417	651
Contig N50	2,148,059	3,443,960
Contig L50	85	34
Contig L90	1,033	139
QV	58.6171	64.1921
Kmer compl.	7.43497	78.242
BUSCO sing.	97.3%	97.6%
BUSCO dupl.	2.1%	1.8%
BUSCO frag.	0.1%	0.1%
BUSCO miss.	0.5%	0.5%

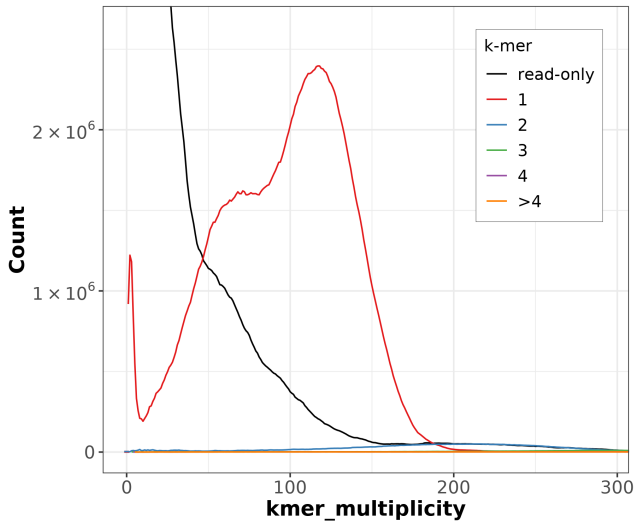
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: arthropoda_odb10 (genomes:90, BUSCOs:1013)

HiC contact map of curated assembly

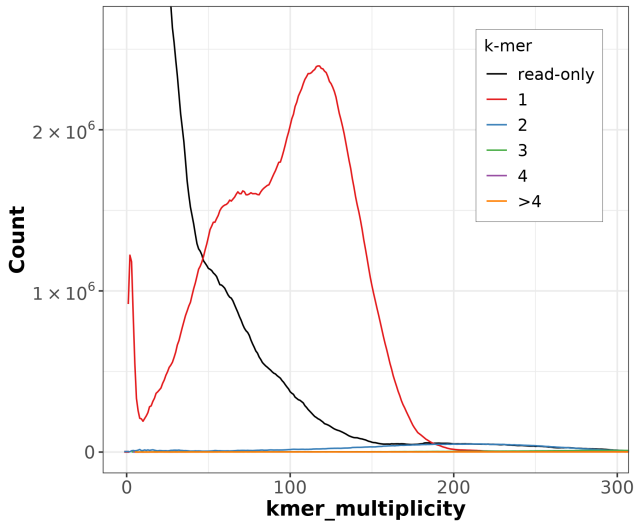
pri HiC PNG is missing!

pri File link is missing!

K-mer spectra of curated assembly



Distribution of k-mer counts per copy numbers found in asm



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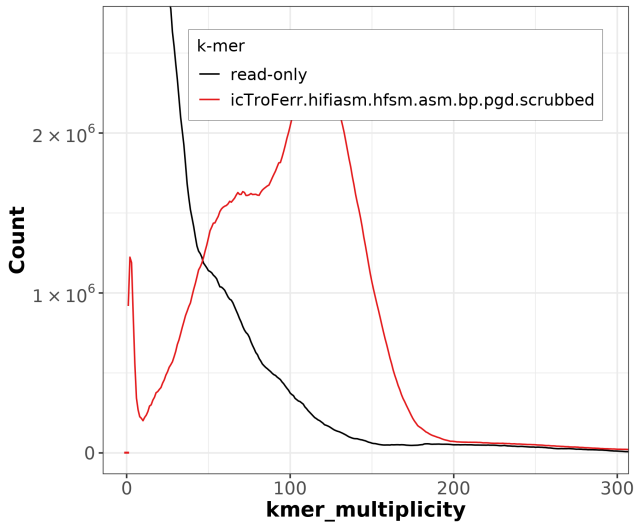

Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening

Data profile

Data	Pacbio HiFi	Hi-C
Coverage	NA	NA

Assembly pipeline

- **CLAWS pipeline**
    |_ *ver:* 3.0
    |_ *key param:* NA
- **Hifiasm**
    |_ *ver:* 0.24.0
    |_ *key param:* --telo-m TTAGG
- **purge_dups**
    |_ *ver:* 1.2.6
    |_ *key param:* NA
- **Merqury**
    |_ *ver:* 1.3
    |_ *key param:* NA
- **BUSCO**
    |_ *ver:* 6.0.0
    |_ *key param:* NA
- **tidk**
    |_ *ver:* 0.2.65
    |_ *key param:* NA
- **chromap**
    |_ *ver:* 0.3.2
    |_ *key param:* NA

Curation pipeline

- **FCS-GX**
    |_ *ver:* 0.5.5
    |_ *key param:* NA

Submitter: Francisco Camara
Affiliation: CNAG

Date and time: 2026-03-04 16:03:21 CET